Signal – Backend Documentation

Pre-requisites:

- Virtual server (VPS) with Ubuntu 24.04 installed
- Root privileges or an account with sudo
- Windows 10 or later or Windows Server 2016 or later
- An Intel Pentium 4 processor or later that is SSE3 capable

Randomly generated base64 strings will be needed for creation of new keys and users. Therefore, the following website can be used to do so:

https://generate.plus/en/base64

Length must equal 4 for any keys.

API Requests/Commands

GET /health:

Checks status of server running.

Command:

curl http://localhost:8080/health

POST /v1/register:

Registers a user using keys.

Command:

curl -X POST http://localhost:8080/v1/register \

```
  -H "Content-Type: application/json" \

  -d
'{"username":"USERNAME","identityKeyPubB64":"PREKEY","signedPreKeyPubB64":"PREKEY","preKeys":[{"keyId":1,"preKeyPubB64":"PREKEY"}]}'
```

POST /v1/login:

Logs in a user using their identity key.

Command:

```
curl -X POST http://localhost:8080/v1/login \

  -H "Content-Type: application/json" \

  -d '{"username":"USERNAME","identityKeyPubB64":"ENCODED TEXT"}'
```

POST /v1/prekeys/refill:

Refills a user's prekeys.

The postgres database needs to be checked before this command is used to ensure a prekey with an unused ID is generated.

Command:

```
curl -X POST http://localhost:8080/v1/prekeys/refill \

  -H "Authorization: Bearer <YOUR TOKEN>" \

  -H "Content-Type: application/json" \

  -d '{"preKeys":[{"keyId":NUMBER,"preKeyPubB64":"PREKEY"}]}'
```

GET /v1/keys/:username:

Retrieves a user's keys to establish communication.

Command:

curl http://localhost:8080/v1/keys/USER

POST /v1/messages:

Sends a user a message using the sender's login token.

Command:

```
curl -X POST http://localhost:8080/v1/messages \
  -H "Authorization: Bearer <YOUR TOKEN>" \
  -H "Content-Type: application/json" \
  -d '{"to":"USERNAME","bodyB64":"ENCODED TEXT"}'
```

GET /v1/messages:

Retrieves all the messages a user has received.

Command:

```
curl http://localhost:8080/v1/messages \
  -H "Authorization: Bearer <YOUR TOKEN>"
```

Other Useful Commands

sudo -u postgres psql sigmsg

Logs into the postgres database where SQL commands can be used to find out information about the backend. The tables include: "users", "prekeys" and "messages".

sudo systemctl daemon-reload

sudo systemctl restart sigserver

Restarts the server.

sudo journalctl -u sigserver -f

Retrieves the last debugging statements recorded regarding the server’s activity.